

PROFILE

BioAI Research Engineer at InstaDeep (BioNTech Group), with three years applying machine learning to clinical trials and protein science. Biological engineer by training, now working where large language models and structural biology meet biomedical data.

EXPERIENCE

BioAI Research Engineer Jun 2023 – Present
InstaDeep (BioNTech Group), Tunis, Tunisia

Clinical Trials: outcome & duration prediction with biomedical LLMs / core contributor

- Built phase-level pipelines for trial outcome prediction on the public TOP benchmark and for trial duration regression, fine-tuning BioBERT (108M) and BioMistral (7B) via full fine-tuning and LoRA on multi-GPU DDP, over features parsed from ClinicalTrials.gov records
- Showed probing frozen encoders matches or exceeds full fine-tuning at a fraction of the trainable parameters; both outperformed HINT, the prior state of the art, across Phase I–III (F1, ROC-AUC, PR-AUC)
- Mentored two interns to delivery on adjacent workstreams: a patient–trial matching model and an LLM assistant for querying trial data

Immunology: TCR–pMHC interface scoring & affinity prediction / core contributor

- Repurposed AlphaFold2 Multimer as a conformational ensemble generator (seed variation, inference-time dropout, retained recycling intermediates), sampling hundreds of models per complex to expose interface uncertainty that ipTM scoring misses
- Built a multi-tool interface scoring pipeline (ModFOLDdock2, VoroMQA, GNINA, Arpeggio) and used stability selection to distil thousands of features into a compact set of robust TCR–pMHC affinity predictors (none of them AlphaFold confidence metrics), attributing the signal via SHAP to interface contact chemistry and CDR loop quality, and validating transferability on an independent de novo binder-design dataset

DeepChain: model applications / owned end-to-end

- Led development and deployment of protein prediction applications on InstaDeep's DeepChain platform (signal peptide and enzyme class prediction), containerised with Docker and run on GCP and Alchor

Bio Skills for Claude Code: internal bioinformatics toolkit / created & led

- Sole author of all six skills in a multi-agent toolkit turning natural-language questions into bioinformatics tool calls: custom subagents per task, agent-to-agent orchestration, Claude Code hooks and plugins, and provenance-tracked pipelines routing across 22 data sources (UniProt, Ensembl, RCSB PDB, AlphaFold, PubMed); in internal use at InstaDeep

DETAILS

MOBILITY
Open to relocation (visa sponsorship required)

LINKEDIN
/in/burak-bakkar

TECHNICAL SKILLS

Machine learning
PyTorch · TensorFlow · Hugging Face Transformers · scikit-learn · LLM fine-tuning, LoRA & probing · distributed training (PyTorch DDP) · multi-agent orchestration · Claude Code skills & hooks · MCP (FastMCP)

Biological data
Protein sequence & structure · AlphaFold outputs · structural confidence metrics · clinical trial registries · enzyme & peptide annotation

Engineering
Python · Docker · Git · GitHub Actions (CI) · GCP · Alchor · MLflow · Neptune.ai · pandas · NumPy · Matplotlib · Seaborn · LaTeX

LANGUAGES

Arabic	Native
French	C2 · near-native
English	C1 · professional
Italian	A1
German	A1 · learning

EDUCATION

Biological Engineer's Degree

Sep 2019 – Nov 2022

National Engineering School of Sfax (ENIS), Sfax, Tunisia

Preparatory Cycle in Biology & Geology

Sep 2016 – Jun 2019

ISEP BG, Tunis, Tunisia

National engineering entrance exam: ranked **63rd of 3,800 candidates** (top 2%) for 350 available places

Scientific Baccalaureate

Jun 2016

High School Rue de Russie, Tunis, Tunisia

EARLIER EXPERIENCE

Science & Robotics Trainer

Nov 2022 – Jun 2023

Play Mind, La Marsa, Tunis, Tunisia

Taught programming and collaborative problem-solving to school-age students.

Graduation Project in Food Process Engineering

Mar – Nov 2022

LGEM Laboratory, ENIS, Sfax, Tunisia

Optimised the production and characterisation of a quinoa-based beverage (*Chenopodium quinoa* Willd.), covering process optimisation, testing and validation, a food application, and a technical-commercial study.

Bioinformatics Intern

Jul – Aug 2021

Centre of Biotechnology of Sfax, Sfax, Tunisia

Screened genomic data for COVID-19 candidate genes.

Quality Control Intern

Jul – Aug 2020

ADWYA Pharmaceuticals, Tunis, Tunisia

HPLC assays, physico-chemical control, and organoleptic verification under pharmaceutical QC protocols.

Molecular Microbiology Intern

Jul – Aug 2019

Pasteur Institute of Tunis, Tunis, Tunisia

Yeast isolation and plating, spot-assay phenotyping, genomic DNA extraction and molecular identification.

LEADERSHIP

President

Biological Engineering Club, ENIS (2019–2021)

Sponsorship Manager

ENACTUS ENIS (2020–2021)

Member

ENIS Debate Club (2019–2020)

President

Drama Club, ISEP-BG (2018–2019)

INTERESTS

Theatre · writing & poetry · music · gaming · volleyball · travel